

Multiple Choice (10 marks)

1. A _____ is a sequential segment of the memory location that is allocated for containing some data such as a character string or an array of integers.
 - (a) stack
 - (b) queue
 - (c) external storage
 - (d) buffer
2. Old operating systems like _____ and NT-based systems have buffer-overflow attack a common vulnerability.
 - (a) Windows 7
 - (b) Chrome
 - (c) IOS 12
 - (d) UNIX
3. Which of the following are most vulnerable to injection attacks?
 - (a) Session IDs
 - (b) Registry keys
 - (c) Network communications
 - (d) SQL queries based on user input
4. In a _____ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
 - (a) Phishing
 - (b) MiTM
 - (c) Buffer-overflow
 - (d) Clickjacking
5. A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
 - (a) front-door
 - (b) backdoor
 - (c) clickjacking
 - (d) key-logging

True / False (10 marks)

Answer True or False

6. True or False: Availability is not considered one of the classic security properties in cybersecurity, which include confidentiality, integrity, and correctness.
7. True or False: A proxy firewall filters traffic at the application layer, inspecting and managing data at a higher level than other firewalls.
8. True or False: DaCryptic is known as a remote Trojan that primarily targets mobile devices for data theft.
9. True or False: Message confidentiality means that the data must arrive at the receiver exactly as they were sent, protecting it from unauthorized access.
10. True or False: Eavesdropping is considered a legitimate method for securing network communications.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the count of divisors is even or odd.
12. Write a python function to count the maximum number of equilateral triangles that can be formed within a given equilateral triangle.

End of Paper